

CHEN, YIXIANG

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🎓 EDUCATION

The Hong Kong Polytechnic University & University of Macau 2026 – Present

Ph.D. in the [Dual Doctoral Degree Programme in Robotics and Embodied Intelligence](#)

- Jointly offered by The Hong Kong Polytechnic University (PolyU) and University of Macau (UM)
- Supervised by [Prof. Wanyu Lin](#) at PolyU and [Prof. Li Li](#) at UM

The Hong Kong University of Science and Technology (HKUST), Hong Kong 2022 – 2026

B.Sc. in Data Science and Technology (DSCT), June 2026

- S.S. Chern Class Student (2023 – 2026)
- CGA: 3.6 / 4.3 *First Class Honors*

Ecole Polytechnique Fédérale de Lausanne (EPFL), Lausanne, Switzerland 2024 – 2025

Exchange

- School of Computer and Communication Sciences IC

High School Affiliated to Shanghai Jiao Tong University, Shanghai, China 2019 – 2022

High School Diploma

✍ PUBLICATION

The Curse of CoT: On the Limitations of Chain-of-Thought in In-Context Learning.

Tianshi Zheng*, [Yixiang Chen*](#), Chengxi Li*, Chunyang Li, Qing Zong, Haochen Shi, Baixuan Xu, Yangqiu Song, Ginny Y Wong, Simon See.

Accepted to Transactions on Machine Learning Research (TMLR), 2025. [\[pdf\]](#), [\[code\]](#)

Self-Redraft: Eliciting Intrinsic Exploration-Exploitation Balance in Test-Time Scaling for Code Generation.

[Yixiang Chen*](#), Tianshi Zheng*, Shijue Huang, Zhitao He, Yi R. Fung.

Preprint, 2025. [\[pdf\]](#)

👤 RESEARCH EXPERIENCE

I am generally interested in Natural Language Processing. More specifically, I focus on advancing the **logical reasoning** and **multi-agent collaboration** capabilities of Large Language Models. In Spring 2025, I co-authored a paper [The Curse of CoT](#) on LLM logical reasoning (accepted to TMLR) with [Tianshi Zheng](#) under the supervision of [Prof. Yangqiu Song](#).

Large Language Models as Autonomous Agents 2025–26 Fall

Undergraduate Researcher Supervisor: Prof. Yangqiu Song

- Explore the enhancement of **LLM-powered agents for scientific discovery**.

Human-Centric Trustworthy AI4Coding 2025–2026

Final Year Thesis Supervisor: Prof. Yi R. Fung

- Complete the paper *SELF-REDRAFT: Eliciting Intrinsic Exploration-Exploitation Balance in Test-Time Scaling for Code Generation*, which is currently under review.
- Gain hands-on experience in **supervised fine-tuning** and **reinforcement learning**.
- Establish a baseline for **intrinsic exploration-exploitation balancing** in **test-time scaling** and identifies feedback and discrimination as key areas with potential for future advances.

Logical Inference and Rationales in Large Foundation Models, HKUST

2024–25 Spring

Undergraduate Researcher Supervisor: Prof. Yangqiu Song

- Co-author the paper *The Curse of CoT: On the Limitations of Chain-of-Thought in In-Context Learning* on LLM logical reasoning (**accepted to TMLR**).
- Investigate the counter-intuitive phenomenon where Large Language Models underperform in a **Chain-of-Thought** setting compared to direct answering during **in-context learning**.

Large-Scale Spatiotemporal Data Analytics and Learning, HKUST

2023–24 Summer

Undergraduate Research Opportunities Program Supervisor: Prof. Xiaofang Zhou

Performance Analysis of PointNetLK for Point Cloud Registration Across Diverse Datasets

ACADEMIC HIGHLIGHTS

In addition to my research experience, I have an excellent academic record, particularly in **statistics** and **mathematics related to machine learning**. Below is a selection of relevant courses where I achieved high grades:

- **Courses with Grade A+:**
 - Statistical Inference
 - Applied Statistics
 - Regression Analysis
 - Foundation of Mathematics
- **Courses with Grade A:**
 - Stochastic Process
 - Honors Probability
 - Matrix Computation
 - Programming with C++
 - Honors Discrete Math Tools for CS
 - Multivariable Calculus
 - Game Theory
- Completed the **Machine Learning** course (CS-433), a core component of the CS master's program at EPFL, with an excellent score of **5.5/6.0**.

★ HONORS AND AWARDS

HKSAR Government Scholarship Fund – Reaching Out Award (ROA) , Education Bureau	2024–2025 Fall
University's Scholarship Scheme for Continuing Undergraduate Students , HKUST	Dec. 2024
Chern Class Scholarship , HKUST	Aug. 2024
Dean's List , HKUST	2023–24 Spring
Dean's List , HKUST	2023–24 Fall
Chern Class Scholarship , HKUST	Aug. 2023
Dean's List , HKUST	2022–23 Spring

SKILLS

- Natural Language Processing / Machine Learning: Supervised Fine-tuning, Reinforcement Learning.
- Programming Languages: Python, C++, SQL, R.

i MISCELLANEOUS

- Personal Website: <https://github.com/ericyxchen2003>
- Languages: English - Fluent, Mandarin & Wu Chinese - Native speaker
- Amateur Vocalist/Vocal Percussionist & Music Director at **Tunatics A Cappella**